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w2.1- 1.1 - Deciding on a Project Idea

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w2.1- 1.1 - Deciding on a Project Idea.mp4 — Haruna Media Player

Capstone Design Project

- Design and build a more complex system
- Combine several basic elements that we have learned
  - Read sensor inputs, control motors, etc.
- You define your own project
- Record your design process and submit it for a grade
- All projects must include network communication

So all projects need to include
network communication of some sort.

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w2.1- 1.1 - Deciding on a Project Idea.mp4 — Haruna Media Player

Deciding on a Project Idea

- Not too easy
  - Complexity must be significant
  - Must combine **multiple** elements we have learned
- Not too hard
  - Need to be able to complete it in the given time period
- Complexity is in individual elements and their combination
  - Reading a photo sensor is easy
  - Driving a servo is easy
  - Pointing towards a light source is harder

Now, note that when you're reasoning about the complexity of a design idea,

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w2.1- 1.1 - Deciding on a Project Idea.mp4 — Haruna Media Player

Subtitle scale: 0.2

Mechanical and Physical Aspects

- Many design ideas involve mechanical and physical elements
  - Robots, quadcopters, fish feeders, etc.
- Be careful about mechanical components
  - Building physical devices can be hard
  - May need equipment that you cannot access
- Physical effects may be complex
  - Vibration impacts stability
  - How do you choose a material?

remember that especially with mechanical devices you get these other effects that

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65

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Summary

This lecture outlines the requirements and expectations for the **Capstone Design Project** in an Internet of Things (IoT) context. Here is a summary of the key parameters:

1. Core Objectives

- **Design vs. Build:** You are required to document a complete design process. Physically building the device is **optional** (budget-dependent), but if you do build it, you must submit a video of the working prototype.
- **Integration:** The project must combine multiple elements (e.g., sensors, motors, cameras) to create a system more complex than previous class assignments.
- **Networking Requirement:** Every project **must** include some form of network communication. This can be full Internet connectivity or simple point-to-point communication between devices.

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2. Calibrating Complexity

- **The "Goldilocks" Zone:** It shouldn't be too easy (just a minor tweak of a class lab) nor too hard (it must be achievable within **four weeks**).
- **System Synergy:** Remember that complexity is often greater than the sum of its parts. Integrating two simple components (like a photoresistor and a servo) into a functional system (like an automated light-tracking water gun) adds significant design challenges.

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3. Mechanical & Physical Components

- **The Risk:** Since this isn't a mechanical engineering course, building physical frames or moving parts from scratch can introduce unforeseen issues like **vibration** or stability problems.
- **Tool Access:** Consider whether you have the necessary tools (e.g., drill presses vs. hand drills) before committing to a complex physical build.
- **Recommendation:** Use **prefabricated components** or kits (robot frames, generic mounting boards) whenever possible to avoid the pitfalls of custom mechanical fabrication.

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w2.2- 1.2 - The Design Process

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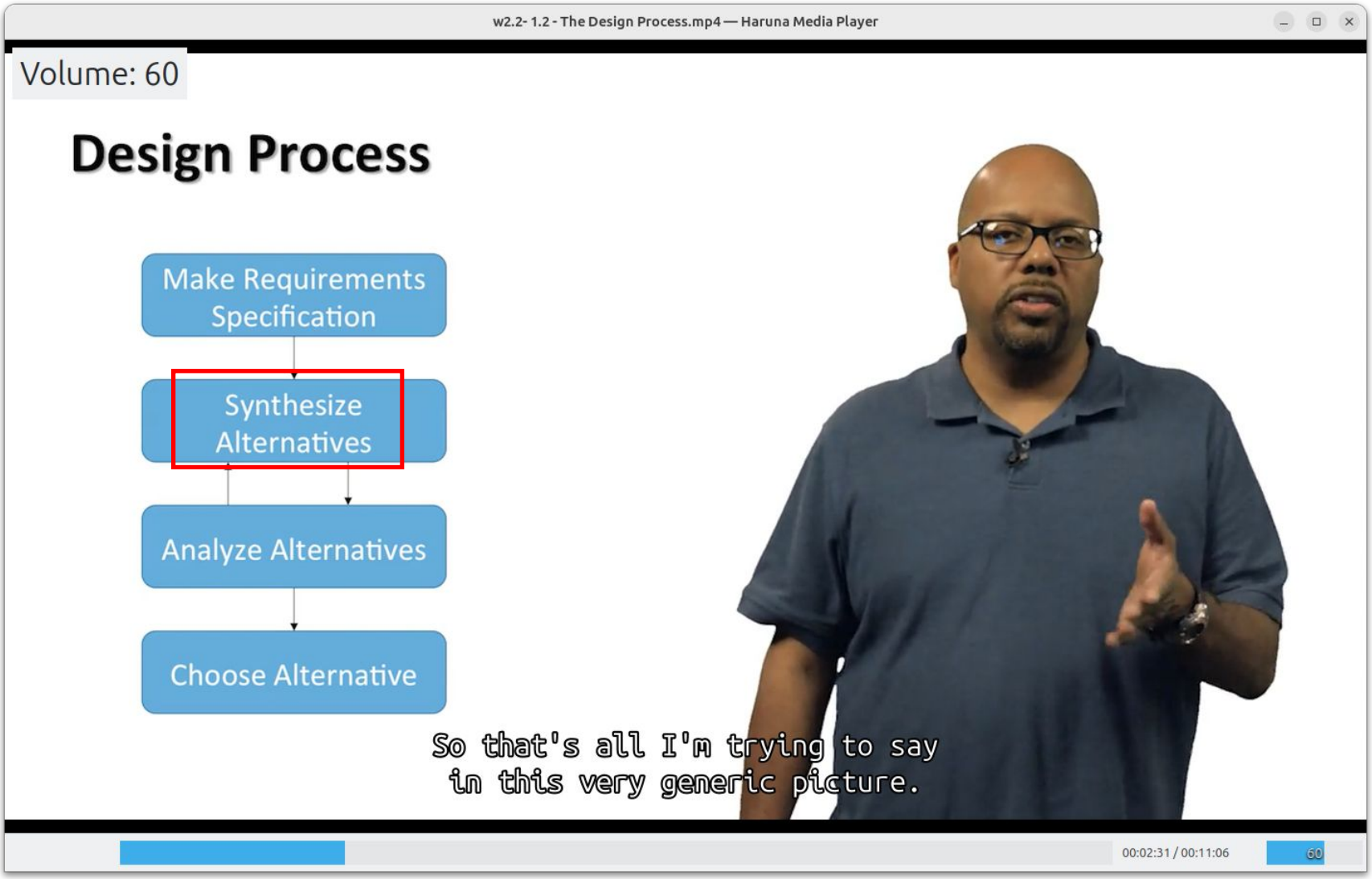

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w2.2- 1.2 - The Design Process.mp4 — Haruna Media Player

Subtitle scale: 0.4

Requirements Specification

- Needs to define the **behavior** of the system
- Not implementation
- Must be understandable by a **user**
  - Should describe how to use the device
- Must be understandable by a **designer**
  - Should completely describe behavior
  - Should describe rare cases as well as common cases

requirement specification will be used
by designers to actually design and

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w2.2- 1.2 - The Design Process.mp4 — Haruna Media Player

Real World Constraints

- Need to consider **non-engineering constraints** on the design
- Existing expertise
- Competing designs and patents
- Financial limitations
- Legal restrictions
  - Quadcopters must be licensed
- Existing design and manufacturing processes

Sometimes you know there are manufacturing processes that you are stuck with.

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Summary

This lecture breaks down how to take an idea from "cool concept" to "professional execution." Here is the breakdown for you:

1. The Generic Design Process

The process follows a logical flow to ensure you don't just "guess" your way through a build:

1. **Requirement Specification:** Write down exactly what the system should do.
2. **Synthesize Alternatives:** Explore different ways to build it (e.g., Raspberry Pi vs. Arduino).
3. **Analyze Alternatives:** Evaluate which option is best based on cost, time, and difficulty.
4. **Choose & Complete:** Pick the winner and finish the actual design/build.

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2. The Requirement Specification (What vs. How)

A good "Spec" defines **Behavior** (what the user sees), not **Implementation** (the code or wires inside).

- **The User View:** Needs to be clear enough for a non-expert to understand what they'll get.
- **The Designer View:** Needs to be fully detailed so the builder knows exactly what happens in every scenario (including rare "edge cases").
- **Intentional Vagueness:** If you don't care what happens when two buttons are pressed at once, say so! This saves the designer from wasting time.

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3. Real-World Constraints

Even a Ninja Architect has to deal with "business" realities. When choosing your design path, consider:

- **Existing Expertise:** It's safer and faster to use what you already know (like the Raspberry Pi we've discussed), though school is a great time to learn one new thing.
- **Financial Limits:** Set a budget and stick to it.
- **Legal/Safety:** For example, in 2026, flying a drone or high-powered rover might require specific licenses or safety checks.
- **Time (Deadlines):** The project must be finished in four weeks, so don't pick a "Board X" that takes three weeks just to ship!

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w2.3- 1.3 - Specification Example

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w2.3- 1.3 - Specification Example.mp4 — Haruna Media Player

Subtitle scale: 0.5

Specification Example: Soda Machine

1. Machine has a coin slot, a soda button, a soda dispensing slot, and a change return slot
2. Coins are inserted into the coin slot
3. If soda button is pressed and enough money has been deposited, then a soda is dispensed and change is returned

✓ Pretty good spec
- Mostly complete
- Describes interface for user
- Some behaviors are not explici

Like for instance,
if you'd ever used a soda machine,

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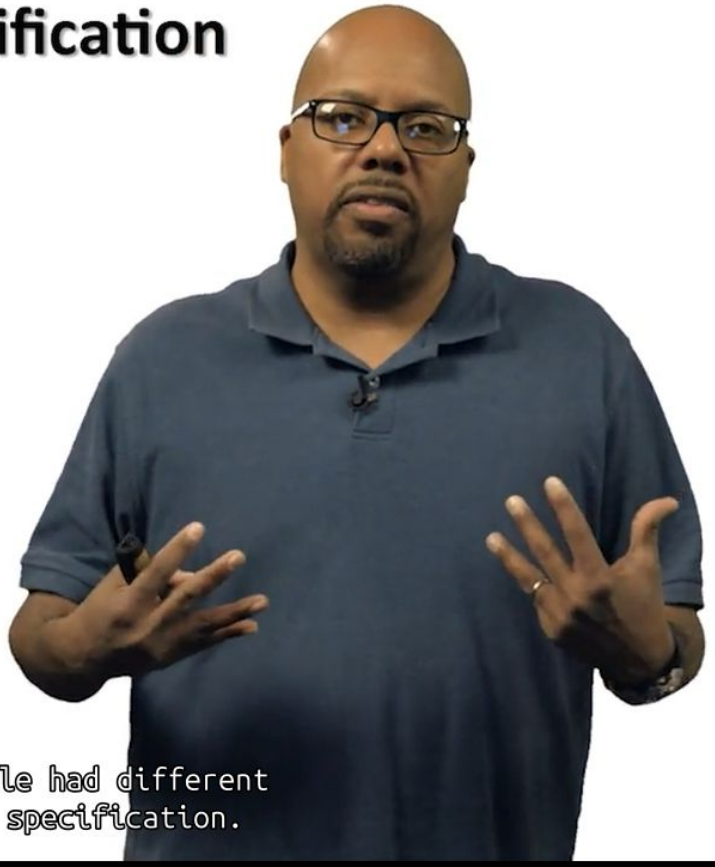
w2.3- 1.3 - Specification Example.mp4 — Haruna Media Player

Another Soda Machine Specification

1. Insert money, press button, get soda

- Less detailed specification
- Some details may be assumed



be trouble because people had different
understandings of the specification.

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w2.3- 1.3 - Specification Example.mp4 — Haruna Media Player

Another Soda Machine Specification

1. Insert money, press button, get soda
2. Arduino is connected to a coin sensor to track money
3. Arduino is connected to servo to dispense soda
4. Arduino is connected to coin dispenser to give change

- Over-specified
  - Implementation detail is provided

So this is bad because
it's over-specified.

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Summary

This lecture covers the pitfalls and best practices of writing a **Requirement Specification** using a soda machine as a case study. Here is the summary for your 1981 Master Ninja Architect toolkit:

1. The "Goldilocks" Specification (Good)

A strong specification balances completeness without getting bogged down in the "how."

- **Elements:** Lists the interface components (coin slot, buttons, dispensing slot).
- **Logic:** Defines the core flow (Coins in → Button press → Soda out → Change returned).
- **Visuals:** Includes a picture or diagram to clarify the user interface.
- **Room for Improvement:** Could explicitly state what happens if the "Coin Return" is pressed without buying a soda.

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2. The Under-Specified Trap (Weak)

Example: *"Insert money, press button, get soda."*

- **The Risk:** It assumes **background knowledge**. You assume everyone knows how a soda machine works, but behavior varies globally (e.g., the lecturer's example of traffic lights in Shanghai vs. the US).
- **The Result:** If the designer and the user have different assumptions about "standard" behavior, the final product will fail.

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3. The Over-Specified Trap (Weak)

Example: *"Use an Arduino to connect to a coin sensor and a servo to dispense soda."*

- **The Mistake:** Including **Implementation Details**.
- **The Result:** This locks the designer into a specific technology (Arduino) too early. A requirement specification should only say **what** the machine does, leaving the choice of **how** (Arduino vs. Raspberry Pi) to the design phase.